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| **PI Name:** | Kennedy, Ann R. D.Sc. | | |
| **Project Title:** | Mechanisms by which Selenomethionine Protects Against Space Radiation Biological Effects | | |
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| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
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| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
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This research program is focused on the mechanism(s) by which L-selenomethionine reduces the levels of biological effects associated with the exposure of cells to high-energy, highly charged particles known as HZE particles from the new NASA BAF facility at the Brookhaven National Laboratory. It is known that L-selenomethionine can reduce the levels of oxidative stress produced by cellular irradiation with x-rays, protons and HZE particles. It is expected that L-selenomethionine will serve as an effective countermeasure for other types of space radiation induced biological effects, such as the induction of cancer. The studies being performed as part of this grant are exploring the mechanisms by which L-selenomethionine can serve as a countermeasure for radiation effects, including the mechanisms by which it may regulate gene expression related to radiation induced carcinogenesis. This research project is driven by our hypothesis that radiation is capable of initiating an ongoing cellular process (by a high frequency, epigenetic event) that is characterized by a state of genetic instability; a later event or events, presumably genetic, then lead directly to

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| **Task Description:** | malignant transformation. The assumptions are that the heritable epigenetic event is a change in gene expression which is induced by the radiation exposure and that L-selenomethionine is capable of preventing, suppressing or reversing this change in gene expression. Microarray technology is being used to assess the effects of SeM on HZE particle radiation-induced changes in gene expression. These studies utilize both in vitro and in vivo model systems. Studies on malignant transformation in vitro utilize HTori-3 cells, which represent a human thyroid epithelial cell model system, and studies on gene expression and function utilize HTori-3 cells and Sprague-Dawley rats as the model systems. Studies with x-rays are also being performed so that Relative Biolgoical Effectiveness (RBE) values for the HZE particles can be determined for radiation induced malignant transformation. By studying the mechanisms by which selenomethionine protects against space radiation induced biological effects, information will be obtained relevant to the mechanisms involved in HZE particle induced radiation carcinogenesis. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Knowledge of the mechanisms by which selenomethionine serves as a countermeasure for radiation induced biological effects could help to develop more effective ways to prevent radiation induced biological damage leading to the known biological effects of radiation (e.g., cell and organ toxicity, mutation and malignant transformation). Research on the mechanisms involved in the prevention of radiation induced biological effects will also have earth benefits, as life on Earth involves continual exposure to various types of radiations which are capable of producing adverse biological effects. |
| **Task Progress:** | As described above, our hypothesis is that SeM prevents or suppresses biological effects in cells exposed to HZE particle radiation by altering gene expression patterns in irradiated cells. Our data clearly show the effectiveness of SeM as a countermeasure for HZE particle radiation-induced differential gene expression. For two radiation doses used, a non-toxic 10 cGy and a slightly toxic 20 cGy dose of radiation from iron ions, supplementation with SeM leads to a clear change in the pattern of HZE particle radiation-induced differential gene expression. In HTori-3 cells exposed to a 10 cGy radiation dose from iron ions, 218 genes are up-regulated and 82 genes are down-regulated in the absence of SeM; in cells irradiated with the same dose of radiation in the presence of SeM, only 75 out of 218 up-regulated and 15 out of 82 down-regulated genes are still differentially expressed. In HTori-3 cells exposed to a 20 cGy dose of radiation from iron ions, 176 genes are up-regulated and 124 genes are down-regulated in the absence of SeM; in cells exposed to the same dose of radiation in the presence of SeM, only 53 out of 176 up-regulated and 9 out of 124 down-regulated genes are still differentially expressed. Further, a large number of genes were differentially expressed in HTori-3 cells irradiated in the presence of SeM whose expression was not differentially regulated when cells were irradiated, for both radiation doses, in the absence of SeM.<br>During the past funding period of this grant, there have been many studies designed to determine whether the changes in gene expression identified by our previous microarray studies could be confirmed by Real Time PCR analyses. For these studies the RNA isolated for microarray analysis was used for the RT-PCR analysis. cDNA was synthesized via reverse transcription using SuperScript II kit and oligo(dT)12-18 as a primer (both from Invitrogen). The target gene transcripts were quantified by Real Time PCR using forward primers, reverse primers and probes, which were synthesized based on the published nucleotide sequences of the genes identified by microarray analysis. Targets were amplified utilizing AmpliTaq Gold (Roche) in Buffer II and 5 mM MgCl2, through 42 two step cycles (95 degrees C for 15 sec and 60 degrees C for 1 min) in an ABI PRISM 7700 Sequence Detection System. Standard curves were generated for each target and used for target quantity determination in the unknown samples (in triplicates). The mRNA levels of the target genes were normalized to the quantity of GAPDH mRNA. HTori-3 cells, a human thyroid epithelial cell line, were maintained in T-25 culture flasks (Falcon) for 3 days prior to irradiation. The medium was either supplemented with 5 micromolar L-selenomethionine or it was supplement free. At the time of the radiation exposure, the cells were confluent. Exposure to HZE particles was performed at the NASA Space Radiation Laboratory (NSRL) Facility at the Brookhaven National Laboratory. Cells were exposed to 0, 10 or 20 cGy of 1.0 GeV/amu iron ions. Six and sixteen hours post-irradiation, cells were harvested for RNA isolation. Cells were frozen in RNA preserving solution (RNA Later, Quiagen) and transported to the University of Pennsylvania, where further steps for isolation of total RNA were taken using RNeasy kit, and according to the manufacturer's instructions.<br><br>Based on our findings using microarray technology, seventeen genes of interest were chosen for validation via quantitative PCR. In alphabetical order these were: Transcription factor AP-1, B-cell CLL/lymphoma 2 (Bcl-2), bone morphogenetic protein 2 (BMP2), bone morphogenetic protein 6 (BMP6), Caspase 1 (CASP1), Collagen 1A1 (COL1A1), Early Growth Response 1 (EGR1), E1A binding protein p300 (EP300), Growth Arrest and DNA Damage Induced 45A (GADD45A), Misshapen – NIK Kinase (MINK), Protein Kinase C-H, Selenoprotein W, 14-3-3 protein sigma (Stratifin - SFN), Mothers against decapentaplegic homolog 3 (SMAD3), Transforming Growth Factor Beta (TGFBeta), Thyoredoxin reductase 3 (TRXRD3) and WW domain containing oxidoreductase (WWOX). In our studies using quantitative PCR, 9 out of 17 genes of interest (about 53%) were partially or totally confirmed to be regulated as the microarray analysis had suggested. The confirmation rate that we have found in our study is similar to the confirmation rates other researchers have observed in their studies using microarrays. For the most part, genes involved in the regulation of growth and differentiation have been the candidates found to be regulated by radiation and/or SeM using both microarray and Q-PCR analyses. These include Early Growth Response 1 (EGR1), Cellular Protein p300 (EP300), Transforming Growth Factor beta (TGbeta), Bone Morphogenetic Protein 2 (BMP2) and Stratifin (SFN). In addition to these, Selenoprotein W has been shown to be up-regulated with SeM treatment in non-irradiated as well as in irradiated cells; this protein may play an important role in redox-related processes that protect cells from radiation-induced oxidative damage, which could cause further damage to the cells, such as DNA damage and damage to the membrane lipids.<br><br>In the next phase of our studies, we will evaluate cellular effects that follow the transcriptional regulation of the candidate genes identified through microarray analysis and confirmed using Q-PCR. Our goal is to determine the molecular pathways involved in cellular responses to HZE particle radiation, which can be modified or enhanced by SeM. It is hoped that the mechanisms by which L-Selenomethionine protects cells from HZE radiation-induced biological effects can be elucidated from these types of studies. |
| **Bibliography Type:** | Description: (Last Updated: 08/24/2020) |

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| **Articles in Peer-reviewed Journals** | Stewart J, Ware J, Fortina P, Breaux J, Gulati S, Kennedy A. "L-selenomethionine modulates high LET radiation -induced alterations of gene expression in cultured human thyroid cells." Oncology Reports, in press 2006. , Apr-2006 |